

FEB12001X – Midterm 20 September 2023

FEB12001X Applied Microeconomics 23/24 (ESE) · 4 exercises · 10.0 points

1 Question 1

3.0 points · 3 questions

Suppose there are two individuals, Anne and Ben, who both derive utility from the consumption of a public good and from the consumption of a private good. Anne's marginal rate of substitution between the public good and the private good is given by

$$MRS_A = \frac{x_A}{G_A + G_B}$$

where x_A is the number of units of the private good that Anne consumes, G_A is the number of units of the public good bought by Anne, and G_B is the number of units of the public good bought by Ben. Ben's marginal rate of substitution is given by

$$MRS_B = \frac{x_B}{G_A + G_B}$$

where x_B is the number of units of the private good that Ben consumes. Marginal production cost of both goods are constant and equal to their price on the market. The price of the public good equals p_G . The price of the private good equals $p_x = 1$. As a result, the marginal rate of transformation between the public good and the private good equals p_G . Anne and Ben have the same level of income y , which they spend fully on the two goods. Given $p_x = 1$, this implies that $x_A = y - p_G G_A$ and $x_B = y - p_G G_B$.

Text

a Consider the assumptions of this exercise. How can you see in this exercise that the public good is non-rival while the private good is rival?

1.0 point · Open · 2/5 Page

Model answer

Non-rivalness implies that units consumed by one person can also be consumed by other people. In this exercise, the MRS depends on the amount of the public good provided by the other person, but not on the amount of the private good provided by the other person

0 points

Incorrect for the public or the private good. An explanation that revolves around the difference in the marginal cost of providing the good between the public and the private good (pg versus 1) makes the answer incorrect.

+1 point

correct answer

b Determine the socially efficient level of units of the public good $G_A + G_B$.

1.0 point · Open · 1/2 Page

Model answer

Condition: $MRS_A + MRS_B = MRT$ (see lecture 1 / chapter 4). $\frac{y - p_G G_A}{G_A + G_B} + \frac{y - p_G G_B}{G_A + G_B} = p_G \rightarrow \frac{2y - p_G G_A - p_G G_B}{G_A + G_B} = p_G \rightarrow 2y = 2p_G (G_A + G_B) \rightarrow (G_A + G_B) = \frac{y}{p_G}$.

0 points

Only writing down the Samuelson condition ($MRS_A + MRS_B = MRT$) is not enough. Any solution without (explicitly or implicitly) using this condition is wrong.

+1 point

correct approach

Suppose that the public good is provided by the market. Anne and Ben decide independently how many units of the public good to buy (of course taking the presence of the other into account).

Text

c Determine the Nash-equilibrium that arises on the market (in terms of provision of the public good by Anne and by Ben).

1.0 point · Open · 4/5 Page

Model answer

For both persons, it is optimal to set $MRS = \frac{p_G}{p_x}$. This gives $\frac{y - p_G G_A}{G_A + G_B} = p_G \rightarrow G_A = \frac{y - p_G G_B}{2p_G}$ and $\frac{y - p_G G_B}{G_A + G_B} = p_G \rightarrow G_B = \frac{y - p_G G_A}{2p_G}$. These can be seen as best-response functions. Solving these

two equations with two unknowns yields $G_A = \frac{y - p_G \left(\frac{y - p_G G_A}{2p_G} \right)}{2p_G} = \frac{\frac{1}{2}y + \frac{1}{2}p_G G_A}{2p_G}$ so that

$G_A \left(1 - \frac{1}{4} \right) = \frac{y}{4p_G} \rightarrow G_A = \frac{1}{3} \frac{y}{p_G}$ so that $G_B = \frac{1}{3} \frac{y}{p_G}$ and $G_A + G_B = \frac{2}{3} \frac{y}{p_G}$. Beyond the question: note that this is inefficiently low.

0 points

Not both best-response functions derived, or stopped after deriving these best-response functions.

+1 point

Both best-response conditions derived and used in solving for G_A and G_B .

2 Question 2

2.0 points · 2 questions

Consider two neighbours, Scott and Jacinda, who live in a country where people are allowed to build campfires in their back yard. If Scott builds a campfire in his back yard, smoke blows to Jacinda's back yard. Jacinda can choose to sit inside or outside (in her back yard). The table below gives the payoffs (in euro's) for each combination of decisions, where the first entry in each cell is Jacinda's payoff and the second entry Scott's payoff.

Text

		Scott	
		Fire	No fire
Jacinda	Inside	10 , 25	10 , 0
	Outside	5 , 40	50 , 15

Scott and Jacinda maximise their own payoff.

Text

a Explain why the Nash-equilibrium differs from the socially efficient outcome. Use maximally four sentences and use the correct market failure in your explanation.

1.0 point · Open · 2/5 Page

Model answer

The socially efficient outcome is Outside, No fire. However, for Scott Fire is a dominant strategy, such that the Nash equilibrium is Inside, Fire. Scott does not take the negative externality he imposes on Jacinda into account when making a campfire.

+1 point

Socially efficient outcome and Nash equilibrium correctly determined, and correct use of negative external effects.

0 points

Incorrect Nash-evenwicht or socially efficient outcome.

0 points

Incorrect explanation or incorrect / no use of external effects.

b It is likely that both conditions of the Coase Theorem are met in the situation described in this exercise. Explain why in maximally four sentences.

1.0 point · Open · 1/2 Page

Model answer

The condition of assigned property rights is fulfilled, as Scott has the right to decide whether to build a campfire in his back yard. Transaction cost of reaching an agreement are probably low, as there are only two individuals involved and it is relatively easy to verify whether Scott sticks to the agreement made.

+1 point

Both conditions mentioned, and low transaction cost linked to few individuals involved and / or easily verified agreement.

0 points

Not both conditions correctly mentioned.

3 Question 3

5.0 points · 4 questions

Company Z employs one worker. The production is described by the following function: $Q = E + K$, where $0 \leq K \leq 1$ and E is the effort chosen by the worker. The price of the product is equal to P . Note that K is an amount of production that is realized independent of the effort of the worker. We assume that $P > 2K$. The worker's utility function is $U = Y - E^2$ and the income Y of the worker is $Y = bQ$. Note that, by assumption, there is no base salary. When the worker chooses not to work for this company, his utility is equal to zero.

Text

a Derive the profit-maximizing bonus b . Do not only provide calculations, but also write a few sentences along the way explaining what you are doing.

2.0 points · Open · 3/4 Page

+1 point

First derive the worker's effort. Utility is $b(E+K) - E^2$. The first-order condition thus is $b - 2E = 0$. Solving gives $E = 0.5b$.

+1 point

Next consider profits. Profits equal $PQ - Y$. Using the production function $Q = E + K$ and the solution for $E = 0.5b$, we can write profits as $P(0.5b + K) - b(0.5b + K)$. Taking the derivative to b gives $0.5P - b - K = 0$. And so $b = 0.5P - K$.

b Explain in a few clear sentences why the level of the profit-maximizing bonus decreases with K .

1.0 point · Open · 1/4 Page

+1 point

It becomes more costly to increase bonus pay when K is higher, because the bonus applied to all output, both output related to E but also output that follows from K .

Assume from now on that, in addition to the bonus b , the company also pays a base salary, denoted by a . Hence, the employment contract now is: $Y = a + bQ$. The base salary can be negative.

Text

c Show that the base salary a that is needed to attract the worker is equal to $-Kb - (b^2)/4$.

1.0 point · Open · 1/2 Page

+1 point

The outside option is zero and so the participation constraint is $a + b(E+K) - E^2 = 0$. Substituting the solution for E and rewriting gives $a = -Kb - (b^2)/4$.

If you made a calculation error, no worries, you get full points if your approach is correct!

d Explain in a few clear sentences why the addition of a base salary to the contract increases the profit-maximizing bonus b . You can derive the profit-maximizing bonus to discover the reason, but a derivation is not necessary for the answer to be correct.

1.0 point · Open · 1/2 Page

+1 point

The base salary enables the company to extract rents from the worker. For instance, the bonus pay that is paid for the K units of production that are not due to effort of the worker can be recouped by reducing a . This can also be seen when writing down profits: $PQ - Y = P(K+E) - A - B(K+E) = P(K+0.5b) + Kb + (b^2)/4 - b(K+0.5b)$. It is easy to see that the second term Kb and the first part of the fourth term $-bK$ cancel out. Hence, the higher cost of increasing the bonus b due to K are nullified by the reduction in the base salary when the bonus rises.

4 Additional answer space

0.0 points · 5 questions

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found.**

Otherwise, the answers in the additional space below will not be reviewed.

NB: Do **not** use this additional space for answers that are too long.

Text

a Additional answer box 1

Open · 4/5 Page

b Additional answer box 2

Open · 1/2 Page

c Additional answer box 3

Open · 1/2 Page

d Additional answer box 4

Open · 1/2 Page

e Additional answer box 5

Open · 1/2 Page